

Practical – Week2

1. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```
ajithmutyala@Ajith:~$ nano filecopy.c
ajithmutyala@Ajith:~$ echo "This is the content of the source file." > source.txt
ajithmutyala@Ajith:~$ cat source.txt
This is the content of the source file.
ajithmutyala@Ajith:~$ gcc filecopy.c -o filecopy
ajithmutyala@Ajith:~$ ./filecopy
File copied successfully.
ajithmutyala@Ajith:~$ cat destination.txt
This is the content of the source file.
ajithmutyala@Ajith:~$ ls -l source.txt destination.txt
-rw-r--r-- 1 ajithmutyala ajithmutyala 40 Aug 14 10:05 destination.txt
-rw-r--r-- 1 ajithmutyala ajithmutyala 40 Aug 14 10:05 source.txt
ajithmutyala@Ajith:~$ diff source.txt destination.txt
ajithmutyala@Ajith:~$ _
```

Report: I compiled and ran `filecopy.c` to copy `source.txt` to `destination.txt`. The program reported "File copied successfully," and `cat destination.txt` showed the content matched the source exactly. `ls -l` confirmed both files were the same size (40 bytes), and `diff source.txt destination.txt` produced no output, indicating no differences. This confirms the program correctly reads from the source file and writes an identical copy to the destination file.

2. Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved

```

ajithmutyala@Ajith:~$ cat sample.txt
Hello Linux System Calls
ajithmutyala@Ajith:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffc8b9a6738 /* 26 vars */)
= 0
brk(NULL)                                = 0x62c2b6c4c000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x7ed7fe2344000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directo
ry)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20943, ...}) = 0
mmap(NULL, 20943, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7ed7fe22e000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...
, 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7ed7fe1fb
000
mmap(0x7ed7fe202000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_
DENYWRITE, 3, 0x7000) = 0x7ed7fe202000
mmap(0x7ed7fe223000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x28000) = 0x7ed7fe223000
mmap(0x7ed7fe22a000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D
ENYWRITE, 3, 0x2f000) = 0x7ed7fe22a000
mmap(0x7ed7fe22c000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_A
NONYMOUS, -1, 0) = 0x7ed7fe22c000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEX
EC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...
, 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7ed7fe1cd
000
mmap(0x7ed7fe1d1000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_
DENYWRITE, 3, 0x4000) = 0x7ed7fe1d1000

```

```

read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7ed7fde45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7ed7fde45cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7ed7fde45cb0}, {sa_handler=0x62c28287f0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7ed7fde45cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7ed7fde45cb0}, {sa_handler=0x62c28287f0c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7ed7fde45cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x62c2b6c8e000) = 0x62c2b6c8e000
statx(AT_FDCWD, "sample.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=25, ...}) = 0
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=25, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 25
splice(4, NULL, 1, NULL, 25, 0Hello Linux System Calls) = 25
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7ed7fe231000, 12288) = 0

```

Report: I ran strace cat sample.txt. The trace showed openat() to open the file, fstat() for metadata, splice() to transfer data directly to stdout, and close() to release the descriptor. This confirms cat uses efficient kernel calls like splice() rather than a plain read/write loop.